where $\overline{\alpha_i}(\mathcal{R}) = \frac{1}{|\mathcal{R}|} \sum_{\boldsymbol{R} \in \mathcal{R}} \alpha_i(\boldsymbol{R})$ is the average concentricity of instance i in $\mathcal{R}$. Concentricity inherits from cosine similarity the invariances to orthogonal transformations and isotropic scaling.

Uniformity. *Uniformity* [153, 52] quantifies density of representations by measuring how close the distribution of instance representations is to a uniform distribution on the unit hypersphere. This measure is defined as

$$m_{\text{uniformity}}(\boldsymbol{R}) = \log \left(\frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} e^{-t \|\boldsymbol{R}_i - \boldsymbol{R}_j\|_2^2} \right), \tag{46}$$

where t is a hyperparameter that was set to $t = 2$ by Wang et al. [151] and Gwilliam and Shrivastava [52]. The statistic is bounded in the interval $[0, 1]$, with $m_{\text{uniformity}}(\boldsymbol{R}) = 1$ indicating perfectly uniform representations. Uniformity is invariant to orthogonal transformations and translation, as these transformations preserve distances.

Tolerance. This statistic considers the proximity of representations of semantically similar inputs [150]. In contrast to the previous statistics, it requires a vector of ground-truth labels $\boldsymbol{y} \in \mathbb{R}^N$. Further, it is assumed that all instance representations have unit norm. *Tolerance* is computed as the mean similarity of inputs with the same class:

$$m_{\text{tol}}(\boldsymbol{R}) = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} (\boldsymbol{R}_i^\top \boldsymbol{R}_j) \cdot \mathbb{1}\{\boldsymbol{y}_i = \boldsymbol{y}_j\}. \tag{47}$$

Tolerance is bounded in the interval $[-1, 1]$, with $m_{\text{tol}}(\boldsymbol{R}) = 0$ indicating that representations that share the same label are always uncorrelated. This statistic is invariant to orthogonal transformations and isotropic scaling.

Instance-Graph Modularity. Similar to the topology-based measures (see Section 3.5), Saini et al. [122] and Lu et al. [87] proposed measures based on building a graph to model representations, though this was not motivated from a topological perspective. Specifically, they used *modularity* [106] to identify whether semantically similar inputs are close together in the graph, and consequently, the representation space. In both cases, a sparse graph was constructed. Saini et al. [122] used the affinity matrix resulting from sparse subspace clustering [38] as the adjacency matrix $\boldsymbol{A} \in \mathbb{R}^{N \times N}$, whereas Lu et al. [87] determined the adjacency matrix element-wise via $\boldsymbol{A}_{i,j} = \boldsymbol{S}_{i,j} \cdot \mathbb{1}\{j \in \mathcal{N}_{\boldsymbol{R}}^k(i)\}$, considering a cosine similarity-based RSM $\boldsymbol{S}$. The modularity of the network, and in consequence the statistic for $\boldsymbol{R}$, is then defined as

$$m_{\text{Mod}}(\boldsymbol{R}) = \frac{1}{2W} \sum_{i,j} \left(\boldsymbol{A}_{i,j} - \frac{d_i d_j}{W} \right) \cdot \mathbb{1}\{\boldsymbol{y}_i = \boldsymbol{y}_j\}, \tag{48}$$

where $d_i = \sum_j \boldsymbol{A}_{i,j}$ denotes the effective degree of node v_i, $W = \sum_{i,j} \boldsymbol{A}_{i,j}$ is a normalization factor, and $\boldsymbol{y}$ is the vector of ground-truth labels. The maximum modularity is given by 1, and high modularity implies that nodes of the same label are highly connected with each other, with only few connections to nodes of another label. Both variants are invariant to orthogonal transformations. The variant by Lu et al. [87] is additionally invariant to isotropic scaling.

Neuron-Graph Modularity. Lange et al. [77] also considered modularity as a statistic to characterize representations. However, in their approach, the nodes v_j represented neurons $\boldsymbol{R}_{-,j}$ instead of instance representations $\boldsymbol{R}_i$. To model the similarity of neurons that is needed to construct the graphs, they proposed four different variants of RSMs that either consider pure neuron activations or also gradients with respect to neuron activations. In that latter case, one may consider the modularity based on such RSMs as a hybrid measure of representational and functional characteristics.

Once an RSM $\boldsymbol{S} \in \mathbb{R}^{D \times D}$ has been computed, Lange et al. [77] constructed the adjacency matrix $\boldsymbol{A}$ of $\mathcal{G}(\boldsymbol{R})$ via $\boldsymbol{A}_{i,j} = \boldsymbol{S}_{i,j} \cdot (1 - \mathbb{1}\{i = j\})$. Unlike Lu et al. [87], they did not allocate nodes to clusters based on ground-truth labels, but determined an optimal soft assignment of n clusters that maximizes modularity. Specifically, they tried to find an optimal cluster assignment matrix $\boldsymbol{C} \in \mathbb{R}^{D \times n}$, where each entry $\boldsymbol{C}_{j,k} \in [0, 1]$ determines the assignment of neuron j to cluster k. The number of clusters $n \leq D$ of neuron activations is a parameter that is to be optimized as well. Given a definition of clustering from Girvan and Newman [45], neuron modularity is then defined as

$$m_{\text{nMod}}(\boldsymbol{R}) = \max_{\boldsymbol{C}} \operatorname{tr}(\boldsymbol{C}^\top \widetilde{\boldsymbol{A}} \boldsymbol{C}) - \operatorname{tr}(\boldsymbol{C}^\top \mathbf{1}_D^\top \mathbf{1}_D \widetilde{\boldsymbol{A}} \boldsymbol{C}), \tag{49}$$

where $\widetilde{\boldsymbol{A}} = \frac{1}{\mathbf{1}_D^\top \boldsymbol{A} \mathbf{1}_D} \boldsymbol{A}$ is the normalized adjacency matrix. To determine the cluster assignment $\boldsymbol{C}$, they provided an approximation method based on Newman's modularity maximization algorithm [105]. Generally, m_{nMod} is invariant to permutations, since these effectively only relabel the nodes in the resulting graph.

4 Functional Similarity Measures

Next, we present functional similarity measures. As mentioned in Section 2.2, these measures compare outputs $\boldsymbol{O}, \boldsymbol{O'} \in \mathbb{R}^{N \times C}$, where each element $\boldsymbol{O}_{i,c}$ denotes the probabilities or scores of class c for input $\boldsymbol{X}_i$, and $\arg\max_c \boldsymbol{O}_{i,c} = \hat{c}$ indicates that class $\hat{c}$ is the prediction for input $\boldsymbol{X}_i$. We mainly categorize measures based on the granularity of the model outputs that they require, as illustrated in Figure 4. An overview of all measures is given in Table 2.

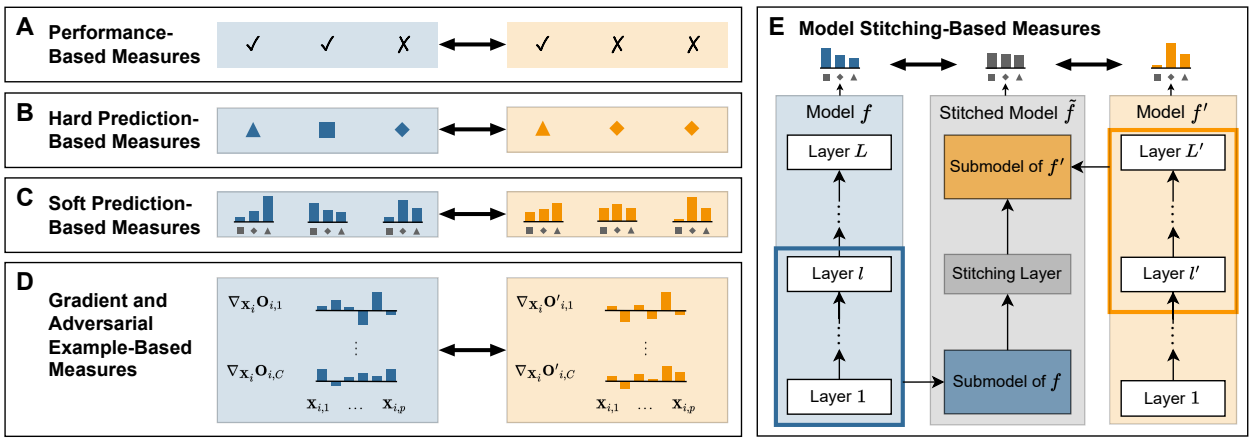

Figure 4: Types of functional similarity measures, illustrated in the context of classifying inputs with respect to their shape ($\diamond, \square, \triangle$). Performance-based (**A**), hard prediction-based (**B**), soft prediction-based (**C**), and gradient and adversarial example-based measures (**D**) compare outputs of different granularity. Model stitching (**E**) combines parts of two models and measures functional similarity between the resulting model and the original models.

Type	Measure	Groupwise	Blackbox Access	Labels Required	Similarity ↑
Performance	Performance Difference	✗	✓	✓	✗
Hard Prediction	Disagreement [90, 39, 86, 126]	✗	✓	✗	✗
	Error-Corrected Disagreement [43]	✗	✓	✓	✗
	Min-Max-normalized Disagreement [69]	✗	✓	✓	✗
	Kappa Statistic [27]	✗*	✓	✗	✓
	Ambiguity [124, 93]	✓	✓	✗	✗
	Discrepancy [93]	✓	✓	✗	✗
	Label Entropy [31]	✓	✓	✗	✗
Soft Prediction	Norm of Soft Prediction Difference [163, 4]	✗	✓	✗	✗
	Surrogate Churn [14]	✗	✓	✗	✗
	Jensen-Shannon Divergence [85]	✗	✓	✗	✗
	Prediction Difference [126]	✓	✓	✗	✗
	Rashomon Capacity [62]	✓	✓	✗	✗
Gradient & Adversarial Ex.	ModelDiff [82]	✗	✗‡	✓	✓
	Adversarial Transferability [63]	✗	✗‡	✓	✓
	Saliency Map Similarity [64]	✗	✗	✗	✓
Stitching	Performance Difference [7, 30, 80]	✗	✗	✓	✗†

*: Groupwise variants available. †: Depends on comparison. ‡: Depends on adversarial example generation.

Table 2: Overview of functional similarity measures. We indicate whether measure enable groupwise comparison of models, whether they can be applied with blackbox access to the models, and if they require ground-truth labels. *Similarity* ↑ indicates whether increasing scores imply increasing similarity of models.

4.1 Performance-Based Measures

A popular view on functional similarity is that models are similar if they reach similar performance on some downstream task (e.g., [34, 80, 30, 7]). This approach is easy to implement, as the comparison of models is reduced to comparing two scalar performance scores, such as accuracy. However, this simplification also obfuscates more nuanced differences in functional behavior, which cannot directly be captured with a single number per model.

Most commonly, given some quality function q that evaluates the performance of a model with respect to ground-truth labels, the (absolute) difference in performance is used for similarity:

$$m_{\text{Perf}}(\boldsymbol{O}, \boldsymbol{O'}) = |q(\boldsymbol{O}) - q(\boldsymbol{O'})|. \tag{50}$$

Although accuracy is an often used quality function in the literature [34, 80, 30, 7], other performance metrics such as F1 score may be used [158]. However, choosing performance metrics that capture relevant aspects of functional behavior requires careful consideration [119].

4.2 Hard Prediction-Based Measures

The measures in this section quantify functional similarity by comparing hard predictions. Thus, each measure of this category will report high similarity if the hard predictions agree for most inputs, regardless of correctness or confidence. These measures are related to literature on ensemble diversity [76, 140] and inter-rater agreement [6, 143].

Disagreement. *Disagreement*, also known as *churn* [39], *jitter* [86], or *Hamming prediction differences* [126], is the expected rate of conflicting hard predictions over inputs and models [131, 90]. Due to its simplicity and interpretability, it is a particularly popular measure for functional similarity. Formally, disagreement between two models is defined as

$$m_{\text{Dis}}(\boldsymbol{O}, \boldsymbol{O'}) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\{\arg\max_j \boldsymbol{O}_{i,j} \neq \arg\max_j \boldsymbol{O'}_{i,j}\}. \tag{51}$$

The measure is bounded in the interval $[0, 1]$, with a score of zero indicating perfect agreement, and a score of one indicating completely distinct functional behavior. Practically, this range is bounded by model quality, with high disagreement being impossible if the compared models are both very accurate. Further, there are bounds on disagreement that depend on the soft predictions of the compared models [14].

Error-Corrected Disagreement. As the range of possible disagreement values depends on the accuracy of the compared models, Fort et al. [43] proposed to correct for this influence by dividing the disagreement by the error rate $q_{\text{Err}}(\boldsymbol{O}) := q_{\text{Err}}(\boldsymbol{O}, \boldsymbol{y}) := \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\{\arg\max_j \boldsymbol{O}_{i,j} \neq \boldsymbol{y}_i\}$ of one of the models:

$$m_{\text{ErrCorrDis}}(\boldsymbol{O}, \boldsymbol{O'}) = \frac{m_{\text{Dis}}(\boldsymbol{O}, \boldsymbol{O'})}{q_{\text{Err}}(\boldsymbol{O})}. \tag{52}$$

By design, this measure is not symmetric since the error rates of the outputs $\boldsymbol{O}, \boldsymbol{O'}$ may vary. A normalized disagreement of zero indicates perfect agreement, whereas the upper limit is dependent on the error rate—exact limits are provided by Fort et al. [43], which help to contextualize the similarity scores that are obtained.

A normalized and symmetric variant of this measure was used by Klabunde and Lemmerich [69]. Their *Min-Max-normalized disagreement* measure relates the observed disagreement $m_{\text{Dis}}(\boldsymbol{O}, \boldsymbol{O'})$ to the minimum and maximum possible disagreement, given error rates $q_{\text{Err}}(\cdot)$ of the models. The minimum is computed as $m_{\text{Dis}}^{(\text{min})}(\boldsymbol{O}, \boldsymbol{O'}) = |q_{\text{Err}}(\boldsymbol{O}) - q_{\text{Err}}(\boldsymbol{O'})|$, and the maximum possible disagreement as $m_{\text{Dis}}^{(\text{max})}(\boldsymbol{O}, \boldsymbol{O'}) = \min(q_{\text{Err}}(\boldsymbol{O}) + q_{\text{Err}}(\boldsymbol{O'}), 1)$, leading to the measure

$$m_{\text{MinMaxNormDis}}(\boldsymbol{O}, \boldsymbol{O'}) = \frac{m_{\text{Dis}}(\boldsymbol{O}, \boldsymbol{O'}) - m_{\text{Dis}}^{(\text{min})}(\boldsymbol{O}, \boldsymbol{O'})}{m_{\text{Dis}}^{(\text{max})}(\boldsymbol{O}, \boldsymbol{O'}) - m_{\text{Dis}}^{(\text{min})}(\boldsymbol{O}, \boldsymbol{O'})}. \tag{53}$$

This measure is bounded in the interval $[0, 1]$, with $m_{\text{MinMaxNormDis}}(\boldsymbol{O}, \boldsymbol{O'}) = 0$ indicating perfect agreement.

Chance-Corrected Disagreement. Rather than correcting for accuracy of models, one can correct for the rate of agreement that two or more classification models are expected to have by chance. The probably most prominent measure that follows this rationale is *Cohen's kappa* [27], which was proposed as a measure for inter-rater agreement, but has also been used in machine learning [22, 44]. Assuming that the compared outputs $\boldsymbol{O}, \boldsymbol{O'}$ are statistically independent, and letting k_c denote the absolute amount of times that class c is predicted in the output $\boldsymbol{O}$, the expected agreement rate of such models is given by $p_e = \frac{1}{N^2} \sum_{c=1}^{C} k_c k'_c$. Based on these values, Cohen's Kappa is defined as

$$m_{\text{Cohen}}(\boldsymbol{O}, \boldsymbol{O'}) = 1 - \frac{m_{\text{Dis}}(\boldsymbol{O}, \boldsymbol{O'})}{1 - p_e} = \frac{p_o - p_e}{1 - p_e}, \tag{54}$$

where $p_o = 1 - m_{\text{Dis}}(\boldsymbol{O}, \boldsymbol{O'})$ denotes the observed agreement. When $m_{\text{Cohen}}(\boldsymbol{O}, \boldsymbol{O'}) = 1$, perfect agreement of the models is indicated; a value $m_{\text{Cohen}}(\boldsymbol{O}, \boldsymbol{O'}) < 0$ indicates less agreement than expected by chance.

To measure similarity between bigger sets of outputs, Fleiss's kappa [42] can be used as a more general variant [36]. The literature on inter-rater agreement [40] lists related measures that are more general or have weaker assumptions.

Groupwise Disagreement. Disagreement cannot identify commonalities across a whole set of models, as pairwise similarity of models does not imply groupwise similarity. The two following measures extend disagreement to identify functional similarity across sets of models.

First, *ambiguity* [93], also called *linear prediction overlap* [52], is the share of instances that receive conflicting predictions by *any* pair of models out of a given set of models. Ambiguity is defined as

$$m_{\text{Ambiguity}}(\mathcal{O}) = \frac{1}{N} \sum_{i=1}^{N} \max_{\substack{\boldsymbol{O}, \boldsymbol{O'} \in \mathcal{O} \\ s.t.\ \boldsymbol{O} \neq \boldsymbol{O'}}} \mathbb{1}\{\arg\max_j \boldsymbol{O}_{i,j} \neq \arg\max_j \boldsymbol{O'}_{i,j}\}. \tag{55}$$

The counterpart to ambiguity is the *stable core* measure proposed by Schumacher et al. [124], which counts the share of instances with consistent predictions. They also considered a relaxation of this consistency, in which an instance is only required to obtain the same prediction by a fixed proportion of models (e.g., 90% of all models) to be considered stable.